

Milestone 3 - Model Adaptation Experiment

Project: NLP-assisted job opportunity matching for MSBA international students

Team GitHub notebook URL: <https://github.com/Kongbai815/job-matching-nlp/blob/main/m3.ipynb>

This notebook compares zero-shot prompting, few-shot prompting, and a true PEFT LoRA transformer on the same four-class job-posting triage task.

1. Data and Experimental Design

- Original public source rows: **785,741**.
- Milestone 2 project sample: **100,000** balanced rows.
- Milestone 3 fixed validation set: **4,000** rows, 1,000 per label.
- Few-shot examples: **64** rows, 16 per label.
- PEFT LoRA training set: **2,000** rows, 500 per label.
- True LoRA base model: **google/bert_uncased_L-4_H-256_A-4**, run on **cpu** for **5** epochs.
- Best LoRA checkpoint: **epoch 4**, selected by validation macro F1.

Code cell 1

```
from pathlib import Path
import json
import pandas as pd

DATA_PATH = Path('jobs.csv')
if not DATA_PATH.exists():
    raise FileNotFoundError('Upload jobs.csv next to this notebook, then run again.')
RESULTS_PATH = Path('outputs/m3.json')
if not RESULTS_PATH.exists():
    raise FileNotFoundError('Keep outputs/m3.json in the cloned repository.')
df = pd.read_csv(DATA_PATH)
results = json.loads(RESULTS_PATH.read_text())
print('Loaded rows:', len(df))
print(pd.crosstab(df['relevance_label'], df['split']))
```

Output

```
Loaded rows: 100000
split      train  validation
relevance_label
high_fit      20000         5000
low_fit       20000         5000
medium_fit    20000         5000
unclear       20000         5000
```

2. Strategy 1 - Zero-Shot Prompting

The zero-shot strategy uses only label definitions. In a hosted LLM setting, this would be a prompt with the four label descriptions and no examples. For a reproducible offline notebook, I implement the same idea as a transparent rubric classifier.

Code cell 2

```
zero = results['strategies']['zero_shot_prompt_rubric']['metrics']
print('Zero-shot accuracy:', round(zero['accuracy'], 4))
print('Zero-shot macro F1:', round(zero['macro_f1'], 4))
```

Output

```
Zero-shot accuracy: 0.8145
Zero-shot macro F1: 0.8147
```

3. Strategy 2 - Few-Shot Prompting

The few-shot strategy adds 16 examples per label. I treat those examples as in-context demonstrations by building class prototypes, which approximates how examples steer a prompt without requiring an external API.

Code cell 3

```
few = results['strategies']['few_shot_prompt_prototypes']['metrics']
print('Few-shot examples:', results['experiment_design']['few_shot_rows'])
print('Few-shot accuracy:', round(few['accuracy'], 4))
print('Few-shot macro F1:', round(few['macro_f1'], 4))
```

Output

```
Few-shot examples: 64
Few-shot accuracy: 0.8285
Few-shot macro F1: 0.8285
```

4. Strategy 3 - True PEFT LoRA Transformer

The third strategy is a full PEFT LoRA experiment using the transformer/PEFT stack. It trains query/value LoRA adapters on a small BERT model while leaving most base-model weights frozen, so it is the most faithful fine-tuning comparison in this milestone.

Code cell 4

```
lora = results['strategies']['true_peft_lora_transformer']
m = lora['metrics']
details = lora['peft_details']
print('Base model:', details['base_model'])
print('PEFT LoRA train rows:', lora['training_rows'])
print('Validation rows:', results['experiment_design']['peft_lora_validation_rows'])
print('Best checkpoint epoch:', details['best_epoch'])
print('LoRA rank:', details['lora']['r'])
print('Runtime seconds:', details['runtime_seconds'])
print('LoRA accuracy:', round(m['accuracy'], 4))
print('LoRA macro F1:', round(m['macro_f1'], 4))
```

Output

```
Base model: google/bert_uncased_L-4_H-256_A-4
PEFT LoRA train rows: 2000
Validation rows: 4000
Best checkpoint epoch: 4
LoRA rank: 8
Runtime seconds: 192.5
LoRA accuracy: 0.8745
LoRA macro F1: 0.8725
```

Adaptation finding. True PEFT LoRA is complete, reproducible, and has the best score in this comparison. The tradeoff is cost and governance: it needs training infrastructure and stronger review because it can learn the weak-label rules too well.

5. Results

Strategy	Adaptation data	Accuracy	Macro F1	Relative cost / effort
Zero-shot prompt rubric	0	0.815	0.815	Lowest - no training examples
Few-shot prototypes	64	0.829	0.829	Low - 64 examples
True PEFT LoRA transformer	2,000	0.875	0.873	High - transformer adapter training

6. 500-Word Analysis

For Milestone 3, I compared three adaptation strategies on the same MSBA job-posting triage task: zero-shot prompting, few-shot prompting, and true PEFT LoRA fine-tuning. The task is to classify real public job postings into high_fit, medium_fit, low_fit, or unclear for a graduate career advisor. I kept the Milestone 2 data plan: the source dataset has 785,741 records and the project sample has 100,000 balanced rows. For this experiment, every strategy was evaluated on the same 4,000-row validation set, with 1,000 examples per label. This common split matters because the comparison is about adaptation strategy, not sampling luck.

The zero-shot strategy uses only label definitions, like a prompt that tells the model what each fit level means. It is the cheapest and easiest approach because it does not need labeled examples or training time. Its result, 0.8145 accuracy and 0.8147 macro F1, is useful but should be interpreted carefully. The labels are weak labels created from transparent screening rules, so zero-shot can match rule language while still missing advisor judgment about whether a job is genuinely appropriate for an international MSBA student.

The few-shot strategy adds sixteen examples per label. In the reproducible offline notebook, I implemented this as prototype matching, which approximates how in-context examples steer a prompt. Few-shot improved to 0.8285 accuracy and 0.8285 macro F1 with only 64 examples. This fits the prompting readings because the model behavior changes through instructions and examples rather than parameter updates. It is operationally attractive: a career advisor can inspect or replace examples without retraining a model.

The third strategy is true PEFT LoRA. I installed the transformer/PEFT stack and trained query/value LoRA adapters on google/bert_uncased_L-4_H-256_A-4 using 2,000 training rows. The run used rank 8, alpha 16, dropout 0.05, batch size 16, and five CPU epochs; the best checkpoint was epoch 4. This is the most faithful fine-tuning experiment because it updates model parameters rather than only changing prompts. It achieved 0.8745 accuracy and 0.8725 macro F1, outperforming few-shot by about 4.4 macro-F1 points. The epoch sweep also showed why a single-epoch run was misleading: the adapter needed several passes before the classifier stabilized. The cost is higher: it requires extra dependencies, training time, and version control around checkpoints.

My recommendation is to use PEFT LoRA as the best adaptation strategy for the final project, but deploy it with a transparent fallback and human review. The performance gain matters because advisors need fewer false recommendations when screening large job lists. However, the current labels are still weak labels, not final human judgments, and the public dataset lacks direct CPT, OPT, or sponsorship evidence. For Milestone 4, I would use the LoRA classifier as the scoring component, expose confidence and rationales, and keep few-shot/prototype rules as an interpretable fallback for uncertain postings. The main lesson is that fine-tuning can help, but only when its extra complexity is governed by clear evaluation and review.